ROYAL GLOBAL UNIVERSITY
GUWAHATI

ROYAL SCHOOL OF DESIGN (RSD)

DEPARTMENT OF PRODUCT DESIGN

COURSE STRUCTURE & SYLLABUS

B.Des. in Product Design

(4-YEARS UNDER GRADUATE PROGRAM)

W.E.F AY: 2026 – 27

COURSE STRUCTURE AND SYLLABUS
DEPARTMENT OF PRODUCT DESIGN

B.Des. in Product Design
Programme Structure

B.Des Course Structure (with subject codes)
Course structure and credit summary

Sl. No.	Courses	1st	2nd	3rd	4th	5th	6th	7th	8th	Total Credits
1	Major	6	6	8	12	12	16	8	8	**76**
2	Minor	3	3	4	6	4	4	4	4	**32**
3	Interdisciplinary (IDC)	3	3	3	NA	NA	NA	NA	NA	**9**
4	AEC	2	2	2	2	NA	NA	NA	NA	**8**
5	SEC	3	3	3	NA	NA	NA	NA	NA	**9**
6	VAC	3	3	NA	NA	NA	NA	NA	NA	**6**
7	Swayam Course	3	3	3	3	NA	NA	NA	NA	**12**
8	Internship/Project/Dissertation	NA	NA	NA	NA	4	NA	8	8	**20**
	Total Credits	**23**	**23**	**23**	**23**	**20**	**20**	**20**	**20**	**172**

Note : There is no multiple entry/exit in the 4 year B.Des. in Product Design Undergraduate program and students shall be awarded the degree only on completion of the full four-year programme with the requisite credits.

1st Semester Major (Core)				
Sl. No.	**Subject Code**	**Names of subjects**	**Level**	**C**
1	PRD082M111	Introduction to Design	100	3
2	PRD082M112	Elements & Principles of Design	100	3
Interdisciplinary				
3	IKS992K101	Indian Knowledge System	100	3
Ability Enhancement Course (AEC)				
4	CEN982A101	Communicative English -1(Introduction to Effective Communication)	100	1
5	BHS982A102	Behavioral Science-I	100	1
Skill Enhancement Course (SEC)				
6	PRD082S111	Illustration Technique	100	3
Value Addition Course (VAC)				
7	VAC992V1415	Material Exploration	100	3
SWAYAM COURSE				
8		Understanding Design	100	3
Minor				
9	PRD082N111	Sketching and Rendering	100	3
Total Credit: 23				

2nd Semester Major (Core)				
Sl. No.	**Subject Code**	**Names of subjects**	**Level**	**C**
1	PRD082M201	Engineering Drawing	100	3
2	PRD082M212	Design Thinking	100	3
Interdisciplinary				
3	IDC0721211	Indian Knowledge System - 2	100	3
Ability Enhancement Course (AEC)				
4	CEN982A201	Communicative English – II (Approaches to Verbal and Non-Verbal Communication)	100	1
5	BHS982A204	Behavioral Science-II	100	1
Skill Enhancement Course (SEC)				
6	PRD082S211	Visualization Techniques	100	3
Value Addition Course (VAC)				
7	VAC992V2110	Nature inspired Design	100	3
SWAYAM COURSE				
8		Design Innovation and Technology	100	3
Minor				
9	PRD082N211	Product Visualization	100	3
Total Credit: 23				

3rd Semester Major (Core)				
Sl. No.	**Subject Code**	**Names of subjects**	**Level**	**C**
1	PRD082M301	Material Studies 1	200	4
2	PRD082M312	Human Factors & Ergonomics	200	4
Interdisciplinary				
3	PRD082I311	North East Art and Craft	200	3
Ability Enhancement Course (AEC)				
4	CEN982A301	Fundamentals of Business Communication	200	1
5	BHS982A304	Behvioural Sciences-III	200	1
Skill Enhancement Course (SEC)				
6	PRD082S311	Form Study	200	3
SWAYAM COURSE				
7		Process Equipment Design	200	3
Minor				
8	PRD082N311	Form Generation	200	4
Total Credit: 23				

4th Semester Major (Core)				
Sl. No.	**Subject Code**	**Names of subjects**	**Level**	**C**
1	PRD082M411	Advanced Product Design and CAD Application	200	4
2	PRD082M412	Packaging Design	200	4
3	PRD082M403	Manufacturing Processes - 1	200	4
Ability Enhancement Course (AEC)				
4	CEN982A401	Employability and Communication	200	1
5	BHS982A404	Behavioural Sciences-IV	200	1
SWAYAM COURSE				
6		Augmenting Design Thinking with Human-Computer Interaction	200	3
Minor				
7	PRD082N411	Product Analysis	200	3
8	PRD082N402	Material Studies	200	3
Total Credit: 23				

5th Semester Major (Core)				
Sl. No.	**Subject Code**	**Names of subjects**	**Level**	**C**
1	PRD082M511	User Interface Graphics	300	4
2	PRD082M512	User Experience	300	4
3	PRD082M503	Material Studies - 2	300	4
Minor				
4	PRD082N511	Manufacturing Processes	300	4
Internship				
5	PRD082M521	4 weeks Internship after 4th Semester		
Project				
6	PRD082M521	Sustainable Product Design Project	300	4
Total Credit: 20				

6th Semester Major (Core)				
Sl. No.	**Subject Code**	**Names of subjects**	**Level**	**C**
1	PRD082M601	Manufacturing Processes - 2	300	4
2	PRD082M612	Research Methodology	300	4
3	PRD082M613	Technically Complex Product	300	4
4	PRD082M614	Socio – Cultural Design	300	4
Minor				
5	PRD082N611	Packaging Design	300	4
Total Credit: 20				

7th Semester Major (Core)				
Sl. No.	**Subject Code**	**Names of subjects**	**Level**	**C**
1	PRD082M711	System Design	400	4
2	PRD082M712	Product Detailing	400	4
Minor				
3	PRD082N711	Design for Sustainability	400	4
Internship				
4	PRD082M722	Internship 8 weeks after 6th Semester	400	4
Project				
5	PRD082M721	Research Project	400	4
Total Credit: 20				

8th Semester Major (Core)				
Sl. No.	**Subject Code**	**Names of subjects**	**Level**	**C**
1	PRD082M811	Design for Environmental Awareness	400	4
2	PRD082M812	Portfolio Design	400	4
Minor				
3	PRD082N812	Minor Project	400	4
Dissertation				
4	PRD082M821	Dissertation/Research Project	400	8
Total Credit: 20				

NOTE : There is no multiple entry/exit in the 4 year B.Des. in Product Design Undergraduate program and students are mandatory to do 8 weeks of Summer/Industry Internship.

NOTE : B.Des Minor students will follow the Track of Department of Fashion Design / Communication Design / Interior Design.
Students opting for Minor tracks can select any one of the 4-track which will enable to attain a minor in the relevant domain.

Minor-Track-I (Communication Design-CD)					
Sl. No.	**Semester**	**Subject Code**	**Names of subjects**	**Level**	**Credit**
1	1st Semester	COD082N111	Illustration Technique	100	3
2	2nd Semester	COD082N211	Visualization Techniques and Photography	100	3
3	3rd Semester	COD082N311	History of Indian Visual Craft	200	4
4	4th Semester	COD082N411	Brand Identity Design	200	3
4	4th Semester	COD082N412	Typography	200	3
5	5th Semester	COD082N511	Publication Design	300	4
6	6th Semester	COD082N611	Copywriting	300	4
7	7th Semester	COD082N711	Advertising & Marketing	400	4

8	8th Semester	COD082N812	Minor Project	400	4
Total Credit: 32					

Minor-Track-II (Fashion Design-FD)					
Sl. No.	**Semester**	**Subject Code**	**Names of subjects**	**Level**	**Credit**
1	1st Semester	FAS082N114	Fashion Design and Concept	100	3
2	2nd Semester	FAS082N214	Fashion Styling and Photography	100	3
3	3rd Semester	FAS082N313	Fundamentals of Pattern making and Garment construction	200	4
4	4th Semester	FAS082N415	Sustainable Product Development	200	3
4	4th Semester	FAS082N416	Introduction to Draping	200	3
5	5th Semester	FAS082N514	Fashion Illustrations	300	4
6	6th Semester	FAS082N615	Visual Merchandising	300	4
6	7th Semester	FAS082N515	Design Process and Portfolio	400	4
7	8th Semester	FAS082N812	Design Portfolio & Collection Development	400	4
Total Credit: 32					

Minor Track-III (Interior Design-ID)					
Sl. No.	**Semester**	**Subject Code**	**Names of subjects**	**Level**	**Credit**
1	1st Semester	IDS082N111	Elements and Principles of Design	100	3
2	2nd Semester	IDS082N211	Colour Theory and Application	100	3
3	3rd Semester	IDS082N311	Art appreciation	200	4
4	4th Semester	IDS082N411	Interior Design Representation Techniques	200	3
4	4th Semester	IDS082N412	Material Documentation & Specification	200	3
5	5th Semester	IDS082N511	Interior	300	4

			Landscape		
6	6th Semester	IDS082N641	Indigenous Knowledge System	300	4
6	7th Semester	IDS082N715	Furnishing, finishing and fittings	400	4
7	8th Semester	IDS082N812	Minor Project	400	4
Total Credit: 32					

Minor Track-IV (Product Design-PD)					
Sl. No.	**Semester**	**Subject Code**	**Names of subjects**	**Level**	**Credit**
1	1st Semester	PRD082N111	Sketching and Rendering	100	3
2	2nd Semester	PRD082N211	Product Visualization	100	3
3	3rd Semester	PRD082N311	Form Generation	200	4
4	4th Semester	PRD082N411	Product Analysis	200	3
4	4th Semester	PRD082N402	Material Studies	200	3
5	5th Semester	PRD082N511	Manufacturing Processes	300	4
6	6th Semester	PRD082N611	Packaging Design	300	4
7	7th Semester	PRD082N711	Design for Sustainability	400	4
8	8th Semester	PRD082N812	Minor Project	400	4
Total Credit: 32					

DETAILED SYLLABUS
DEPARTMENT OF PRODUCT DESIGN

1st Semester		
Paper 1 Major (Core)	Introduction to Design L-T-P-C : 1-0-4-3 Credits : 3 Scheme of Evaluation : Studio	Subject Code: PRD082M111 Level : 100

Course Objective: To introduce learners to the fundamentals of design, its history, disciplines and processes, and to build a vocabulary for creative thinking and making.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand the meaning, scope and disciplines of design.	BT1
CO 2	Apply basic design processes to simple problems.	BT2
CO 3	Analyze examples of good design across contexts.	BT3
CO 4	Create simple design outcomes using foundational methods.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	**Topics (if applicable) & Course Contents**	**Periods/ Hours**	
		L	**P**
I	**What is Design** Definition and scope of design Design disciplines and their interrelation Role of the designer in society	4	15
II	**History and Movements** Evolution of design Key movements: Bauhaus, De Stijl, Modernism Indian design heritage	4	15
III	**The Design Process** Design thinking overview Research, ideation, prototyping Iteration and feedback	4	15
IV	**Design in Practice** Case studies Ethics and sustainability in design Introductory project	3	15
	TOTAL	**15**	**60**

Notional Credit Hours for the course: 30 x 3 = 90

Total Credits in the Paper	**Lecture/ Tutorial**	**Studio/Practical**	**Experiential Learning**
3	15 hours	60 hours	8 hours
			Studio Exercises, Design Diary, Presentations

Text Books:

Ambrose, G. & Harris, P. (2009); Basics Design: Design Thinking; AVA Publishing.

Reference Books:

Heller, S. & Vienne, V. (2012); 100 Ideas that Changed Graphic Design; Laurence King.

Reading Materials:

Norman, D. (2013); The Design of Everyday Things; Basic Books.

1st Semester		
Paper 2 Major (Core)	Elements & Principles of Design L-T-P-C : 1-0-4-3 Credits : 3 Scheme of Evaluation : Studio	Subject Code: PRD082M112 Level : 100

Course Objective: To build a thorough understanding of the visual elements and organizing principles of design and to apply them in two- and three-dimensional compositions.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand the elements and principles of design.	BT1
CO 2	Apply elements and principles to create effective compositions.	BT2
CO 3	Analyze compositions in terms of visual organization.	BT3
CO 4	Create balanced, expressive design compositions.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	**Topics (if applicable) & Course Contents**	**Periods/ Hours**	
		L	**P**
I	**Elements of Design** Point, line, shape, form Texture, value, space Colour basics	4	15
II	**Principles of Design** Balance, contrast, emphasis Rhythm, movement, proportion Unity and variety	4	15
III	**Two-Dimensional Composition** Figure-ground Grid and structure Gestalt principles	4	15
IV	**Three-Dimensional Composition** Form and volume Space and mass Model making	3	15
	TOTAL	**15**	**60**

Notional Credit Hours for the course: 30 x 3 = 90

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
3	15 hours	60 hours	8 hours
			Studio Compositions, Model Making, Critiques

Text Books:

Wong, W. (1993); Principles of Form and Design; Wiley.

Reference Books:

Lauer, D.A. & Pentak, S. (2011); Design Basics; Cengage Learning.

1st Semester		
Paper 3 Minor	Sketching and Rendering L-T-P-C : 1-0-4-3 Credits : 3 Scheme of Evaluation : Studio	Subject Code: PRD082N111 Level : 100

Course Objective: To build strong freehand sketching and rendering skills for communicating product ideas quickly and convincingly.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand sketching and rendering fundamentals.	BT1
CO 2	Apply perspective and rendering techniques to products.	BT2
CO 3	Analyze form and proportion through drawing.	BT3
CO 4	Create presentation-quality product renderings.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	**P**
I	**Sketching Basics** Line quality and control Perspective for products Proportion and form	4	15
II	**Form and Volume** Ellipses and cylinders Light and shadow Construction sketching	4	15
III	**Rendering** Marker rendering Material and texture Digital rendering intro	4	15
IV	**Presentation Sketches** Ideation sketches Explanatory sketches Portfolio renders	3	15
	TOTAL	**15**	**60**

Notional Credit Hours for the course: 30 x 3 = 90

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
3	15 hours	60 hours	8 hours
			Sketching Studio, Rendering Lab, Reviews

Text Books:

Eissen, K. & Steur, R. (2007); Sketching: Drawing Techniques for Product Designers; BIS Publishers.

Reference Books:

Robertson, S. & Bertling, T. (2013); How to Render; Design Studio Press.

1st Semester		
Paper 4 Interdisciplinary (IDC)	Indian Knowledge System L-T-P-C : 3-0-0-3 Credits : 3 Scheme of Evaluation : Theory	Subject Code: IKS992K101 Level : 100

Course Objective: To introduce learners to the Indian Knowledge System (IKS) and its systematized disciplines developed in India from ancient times, and their relevance to contemporary design.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand the scope and significance of the Indian Knowledge System.	BT1
CO 2	Apply insights from traditional Indian knowledge to design thinking.	BT2
CO 3	Analyze Indian arts, crafts and sciences as knowledge traditions.	BT3
CO 4	Appreciate the continuity of Indian knowledge in modern practice.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	P
I	**Introduction to IKS** Meaning and scope of IKS Vedas, Vedangas and Upavedas Oral and textual traditions	12	0
II	**Indian Arts and Aesthetics** Shilpa Shastra Natyashastra and rasa theory Traditional crafts	11	0
III	**Science and Technology in India** Vedic mathematics	11	0

	Astronomy and metallurgy Ayurveda		
IV	**IKS and Contemporary Design** Sustainability in tradition Indigenous materials Case studies	11	0
	TOTAL	**45**	**0**

Notional Credit Hours for the course: 30 x 3 = 90

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
3	45 hours	0 hours	8 hours
			Field Visits, Craft Documentation, Seminars

Text Books:

Mahadevan, B., Bhat, V.R. & Nagendra Pavana R.N. (2022); Introduction to Indian Knowledge System; PHI Learning.

Reference Books:

Kapoor, K. & Singh, A.K. (2005); Indian Knowledge Systems; D.K. Printworld.

1st Semester		
Paper 5 Ability Enhancement course (AEC)	Communicative English -1(Introduction to Effective Communication) L-T-P-C : 1-0-0-1 Credits : 1 Scheme of Evaluation : Theory	Subject Code: CEN982A101 Level : 100

Course Objective: To introduce the learner to the fundamentals of effective communication and build a foundation in listening, speaking, reading and writing skills.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand the communication process and its components.	BT1
CO 2	Apply grammar and vocabulary for clear expression.	BT2
CO 3	Analyze spoken and written texts for meaning.	BT3
CO 4	Demonstrate effective listening and speaking skills.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	P
I	**Introduction to Communication** Communication process Barriers to communication Types of communication	4	0
II	**Grammar and Vocabulary**	4	0

	Parts of speech Sentence structure Vocabulary building		
III	**Listening and Speaking** Active listening Pronunciation Everyday conversation	4	0
IV	**Reading and Writing** Reading comprehension Note-making Basic composition	3	0
	TOTAL	**15**	**0**

Notional Credit Hours for the course: 30 x 1 = 30

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
1	15 hours	0 hours	8 hours
			Language Lab, Group Discussions, Presentations

Text Books:

Kumar, S. & Lata, P. (2015); Communication Skills; Oxford University Press.

Reference Books:

Murphy, R. (2019); Essential English Grammar; Cambridge University Press.

1st Semester		
Paper 6 Ability Enhancement course (AEC)	Behavioral Science-I L-T-P-C : 1-0-4-1 Credits : 1 Scheme of Evaluation : Studio	Subject Code: BHS982A102 Level : 100

Course Objective: The objective of Behavioural Science-I is to develop the learner's self-awareness, interpersonal effectiveness and professional behaviour essential for a collaborative design practice.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand the key concepts of Behavioural Science-I.	BT1
CO 2	Apply behavioural skills in academic and studio settings.	BT2
CO 3	Analyze interpersonal and group situations effectively.	BT3
CO 4	Demonstrate professional and ethical conduct.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	P
I	**Understanding Self and Personality**	4	15

	Self-awareness, self-concept and self-esteem Personality theories and traits Perception and attitudes		
II	**Emotional Intelligence** Understanding emotions Self-regulation and empathy Managing stress	4	15
III	**Interpersonal Skills** Building relationships Active listening Assertiveness	4	15
IV	**Motivation and Goal Setting** Theories of motivation SMART goals Time management	3	15
	TOTAL	**15**	**60**

Notional Credit Hours for the course: 30 x 1 = 30

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
1	15 hours	60 hours	8 hours
			Role Plays, Group Activities, Self-assessment, Reflective Journaling

Text Books:

Robbins, S.P. & Judge, T.A. (2019); Organizational Behavior, 18th Ed.; Pearson.

Reference Books:

Goleman, D. (1995); Emotional Intelligence; Bantam Books.

1st Semester		
Paper 7 Skill Enhancement course (SEC)	Illustration Technique L-T-P-C : 1-0-4-3 Credits : 3 Scheme of Evaluation : Studio	Subject Code: PRD082S111 Level : 100

Course Objective: To develop illustration skills across media and to enable learners to communicate ideas, narratives and concepts through effective visual illustration.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand the fundamentals and styles of illustration.	BT1
CO 2	Apply drawing and rendering techniques across media.	BT2
CO 3	Analyze illustration for narrative and communication.	BT3
CO 4	Create original illustrations for design applications.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	P
I	**Foundations of Illustration** Line, tone and form Observation and gesture drawing Perspective basics	4	15
II	**Media and Techniques** Pencil, ink and marker Watercolour and gouache Digital illustration	4	15
III	**Illustration Styles** Realistic to stylized Character and concept Editorial and narrative	4	15
IV	**Applied Illustration** Illustration for print and screen Storyboarding Portfolio piece	3	15
	TOTAL	**15**	**60**

Notional Credit Hours for the course: 30 x 3 = 90

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
3	15 hours	60 hours	8 hours
			Sketchbooks, Studio Assignments, Live Drawing

Text Books:

Zeegen, L. (2012); The Fundamentals of Illustration; Bloomsbury.

Reference Books:

Hall, A. (2011); Illustration; Laurence King Publishing.

1st Semester		
Paper 8 Value Added Course (VAC)	Material Exploration L-T-P-C : 3-0-0-3 Credits : 3 Scheme of Evaluation : Theory	Subject Code: VAC992V1415 Level : 100

Course Objective: To develop a hands-on understanding of materials, their properties and expressive potential through systematic exploration and experimentation.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand the properties and behaviour of diverse materials.	BT1
CO 2	Apply material exploration techniques in design.	BT2
CO 3	Analyze material suitability for form and function.	BT3

CO 4	Create innovative artefacts through material experimentation.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	P
I	**Introduction to Materials** Natural and synthetic materials Material properties Sustainability of materials	12	0
II	**Material Manipulation** Cutting, joining, folding Surface treatments Additive and subtractive processes	11	0
III	**Material and Form** Material-led form generation Texture and finish Structural behaviour	11	0
IV	**Material Project** Material study board Prototype making Documentation	11	0
	TOTAL	**45**	**0**

Notional Credit Hours for the course: 30 x 3 = 90

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
3	45 hours	0 hours	8 hours
			Material Library, Workshop Experiments, Sample Boards

Text Books:

Lefteri, C. (2014); Materials for Design; Laurence King Publishing.

Reference Books:

Ashby, M. & Johnson, K. (2013); Materials and Design; Butterworth-Heinemann.

2nd Semester		
Paper 1 Major (Core)	Engineering Drawing L-T-P-C : 3-0-0-3 Credits : 3 Scheme of Evaluation : Theory	Subject Code: PRD082M201 Level : 100

Course Objective: To develop the ability to read and produce technical engineering drawings using standard conventions essential for product design.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand engineering drawing conventions and standards.	BT1

CO 2	Apply orthographic and isometric projection.	BT2
CO 3	Analyze components through sectional and detail views.	BT3
CO 4	Produce accurate technical drawings.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	P
I	**Drawing Fundamentals** Instruments and standards (BIS) Lines, lettering and dimensioning Scales	12	0
II	**Projection** Orthographic projection Isometric and oblique First and third angle	11	0
III	**Sections and Details** Sectional views Auxiliary views Detail drawings	11	0
IV	**Applied Drawing** Assembly drawings Component drawings CAD introduction	11	0
	TOTAL	**45**	**0**

Notional Credit Hours for the course: 30 x 3 = 90

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
3	45 hours	0 hours	8 hours
			Drawing Studio, CAD Lab, Assignments

Text Books:

Bhatt, N.D. (2018); Engineering Drawing; Charotar Publishing.

Reference Books:

French, T.E. & Vierck, C.J. (1993); Engineering Drawing and Graphic Technology; McGraw Hill.

2nd Semester		
Paper 2 Major (Core)	Design Thinking L-T-P-C : 1-0-4-3 Credits : 3 Scheme of Evaluation : Studio	Subject Code: PRD082M212 Level : 100

Course Objective: To develop cognitive, strategic and practical abilities in learners through the human-centred design thinking approach, enabling them to frame problems, generate ideas and prototype creative solutions.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand the design thinking mindset and its phases.	BT1
CO 2	Apply empathy, ideation and prototyping methods to design problems.	BT2
CO 3	Analyze user needs and reframe problems from insights.	BT3
CO 4	Create and test design solutions through iterative prototyping.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	**P**
I	**Introduction to Design Thinking** Origins and mindset of design thinking The design thinking process (Empathise, Define, Ideate, Prototype, Test) Convergent and divergent thinking	4	15
II	**Empathy and Problem Framing** User research and observation Empathy maps and personas Defining point of view and problem statements	4	15
III	**Ideation** Brainstorming and ideation techniques Mind mapping and SCAMPER Selecting and developing ideas	4	15
IV	**Prototyping and Testing** Rapid and low-fidelity prototyping Storyboarding User testing and iteration	3	15
	TOTAL	**15**	**60**

Notional Credit Hours for the course: 30 x 3 = 90

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
3	15 hours	60 hours	8 hours
			Design Sprints, Group Work, Prototyping, Presentations

Text Books:

Cross, N. (2011); Design Thinking: Understanding How Designers Think and Work; Berg.

Reference Books:

Brown, T. (2009); Change by Design; Harper Business.

Kelley, T. & Kelley, D. (2013); Creative Confidence; Crown Business.

Reading Materials:

Liedtka, J. (2018, Sept-Oct); Why Design Thinking Works; Harvard Business Review.

2nd Semester

Paper 3 Minor	Product Visualization L-T-P-C : 1-0-4-3 Credits : 3 Scheme of Evaluation : Studio	Subject Code: PRD082N211 Level : 100

Course Objective: To enable learners to visualize product concepts through drawing, models and digital tools for effective communication.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand principles of product visualization.	BT1
CO 2	Apply manual and digital visualization methods.	BT2
CO 3	Analyze products through visual representation.	BT3
CO 4	Create compelling product visualizations.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	**P**
I	**Visualization Fundamentals** Observation drawing Proportion and scale Reference study	4	15
II	**Manual Visualization** Perspective rendering Material rendering Exploded views	4	15
III	**Digital Visualization** 3D basics Rendering and lighting Presentation layouts	4	15
IV	**Visualization Project** Concept visualization Board making Presentation	3	15
	TOTAL	**15**	**60**

Notional Credit Hours for the course: 30 x 3 = 90

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
3	15 hours	60 hours	8 hours
			Studio Work, Digital Lab, Reviews

Text Books:

Eissen, K. & Steur, R. (2007); Sketching: Drawing Techniques for Product Designers; BIS Publishers.

Reference Books:

Powell, D. (1990); Presentation Techniques; Little, Brown.

2nd Semester		
Paper 4 IDC	Indian Knowledge System - 2 L-T-P-C : 3-0-0-3 Credits : 3 Scheme of Evaluation : Theory	Subject Code: IDC0721211 Level : 100

Course Objective: To deepen the learner's engagement with the Indian Knowledge System, focusing on its applications in art, design and sustainable practice.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand advanced themes within the Indian Knowledge System.	BT1
CO 2	Apply IKS principles to design problem solving.	BT2
CO 3	Analyze the ecological and cultural wisdom of Indian traditions.	BT3
CO 4	Create design responses informed by indigenous knowledge.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	**P**
I	**Knowledge Traditions** Regional knowledge systems Guru-shishya parampara Community knowledge	12	0
II	**Indian Visual Culture** Iconography and symbolism Folk and tribal art Temple and vernacular architecture	11	0
III	**Ecology and Sustainability** Traditional ecological knowledge Water and material wisdom Circular practices	11	0
IV	**Application in Design** Contemporary reinterpretation Craft-design collaboration Case studies	11	0
	TOTAL	**45**	**0**

Notional Credit Hours for the course: 30 x 3 = 90

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
3	45 hours	0 hours	8 hours
			Craft Cluster Visits, Documentation, Group Projects

Text Books:

Mahadevan, B., Bhat, V.R. & Nagendra Pavana R.N. (2022); Introduction to Indian Knowledge System; PHI Learning.

Reference Books:

Jaitly, J. (2012); Crafts Atlas of India; Niyogi Books.

2nd Semester		
Paper 5 AEC	Communicative English – II (Approaches to Verbal and Non-Verbal Communication) L-T-P-C : 1-0-0-1 Credits : 1 Scheme of Evaluation : Theory	Subject Code: CEN982A201 Level : 100

Course Objective: To develop the learner's ability to communicate effectively through verbal and non-verbal approaches in personal, academic and professional contexts.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand the principles of verbal and non-verbal communication.	BT1
CO 2	Apply appropriate registers of spoken and written English.	BT2
CO 3	Analyze audience and context for effective communication.	BT3
CO 4	Demonstrate confident presentation and interpersonal skills.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	**P**
I	**Verbal Communication** Spoken English and pronunciation Conversation skills Public speaking	4	0
II	**Non-Verbal Communication** Body language Proxemics and paralanguage Visual communication	4	0
III	**Reading and Writing** Comprehension Paragraph and essay writing Email etiquette	4	0
IV	**Presentation Skills** Structuring a talk Visual aids Group discussion	3	0
	TOTAL	**15**	**0**

Notional Credit Hours for the course: 30 x 1 = 30

Total Credits in	Lecture/ Tutorial	Studio/Practical	Experiential Learning

the Paper			
1	15 hours	0 hours	8 hours
			Group Discussions, Presentations, Role Plays

Text Books:

Kumar, S. & Lata, P. (2015); Communication Skills; Oxford University Press.

Reference Books:

Raman, M. & Sharma, S. (2011); Technical Communication; Oxford University Press.

2nd Semester		
Paper 6 AEC	Behavioral Science-II L-T-P-C : 1-0-0-1 Credits : 1 Scheme of Evaluation : Theory	Subject Code: BHS982A204 Level : 100

Course Objective: The objective of Behavioural Science-II is to develop the learner's self-awareness, interpersonal effectiveness and professional behaviour essential for a collaborative design practice.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand the key concepts of Behavioural Science-II.	BT1
CO 2	Apply behavioural skills in academic and studio settings.	BT2
CO 3	Analyze interpersonal and group situations effectively.	BT3
CO 4	Demonstrate professional and ethical conduct.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	**P**
I	**Group Dynamics** Stages of group formation Roles in groups Cohesion and conflict	4	0
II	**Team Building** Characteristics of effective teams Collaboration Trust and accountability	4	0
III	**Communication in Groups** Verbal and non-verbal cues Feedback Barriers to communication	4	0
IV	**Conflict Management** Sources of conflict Resolution styles Negotiation basics	3	0
	TOTAL	**15**	**0**

Notional Credit Hours for the course: 30 x 1 = 30

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
1	15 hours	0 hours	8 hours
			Role Plays, Group Activities, Self-assessment, Reflective Journaling

Text Books:

Robbins, S.P. & Judge, T.A. (2019); Organizational Behavior, 18th Ed.; Pearson.

Reference Books:

Goleman, D. (1995); Emotional Intelligence; Bantam Books.

2nd Semester		
Paper 7 SEC	Visualization Techniques L-T-P-C : 1-0-4-3 Credits : 3 Scheme of Evaluation : Studio	Subject Code: PRD082S211 Level : 100

Course Objective: To equip learners with visualization and rendering skills to represent ideas, objects and spaces convincingly using manual and digital techniques.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand principles of visualization and representation.	BT1
CO 2	Apply sketching, rendering and digital visualization tools.	BT2
CO 3	Analyze visual references for accurate representation.	BT3
CO 4	Create compelling visualizations of design concepts.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	P
I	**Sketching and Drawing** Freehand sketching Perspective and proxy views Quick concept sketches	4	15
II	**Rendering Techniques** Light, shade and shadow Material and texture rendering Colour rendering	4	15
III	**Digital Visualization** Vector and raster basics Digital rendering tools Composition and layout	4	15
IV	**Presentation Visuals** Concept boards Storyboards	3	15

	Portfolio visuals		
	TOTAL	**15**	**60**

Notional Credit Hours for the course: 30 x 3 = 90

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
3	15 hours	60 hours	8 hours
			Studio Rendering, Digital Labs, Reviews

Text Books:

Robertson, S. & Bertling, T. (2013); How to Render; Design Studio Press.

Reference Books:

Powell, D. (1990); Presentation Techniques; Little, Brown.

2nd Semester		
Paper 8 **VAC**	Nature inspired Design L-T-P-C : 1-0-4-3 Credits : 3 Scheme of Evaluation : Studio	Subject Code: VAC992V2110 Level : 100

Course Objective: To sensitize learners to nature as a source of design inspiration and to apply biomimicry and biophilic principles in creative problem solving.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand the principles of nature-inspired and biomimetic design.	BT1
CO 2	Apply patterns and systems from nature to design.	BT2
CO 3	Analyze natural structures for functional insights.	BT3
CO 4	Create design solutions inspired by natural forms and processes.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	P
I	**Nature and Design** Biomimicry and biophilia Patterns in nature Forms, structures, systems	4	15
II	**Learning from Nature** Structural efficiency Material and surface strategies Ecosystems and cycles	4	15
III	**Biomimetic Methods** Function-to-nature mapping Abstraction of principles Case studies	4	15

IV	**Nature-Inspired Project** Concept development Prototyping Presentation	3	15
	TOTAL	**15**	**60**

Notional Credit Hours for the course: 30 x 3 = 90

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
3	15 hours	60 hours	8 hours
			Nature Study, Sketching, Prototyping

Text Books:

Benyus, J.M. (1997); Biomimicry: Innovation Inspired by Nature; William Morrow.

Reference Books:

Pawlyn, M. (2011); Biomimicry in Architecture; RIBA Publishing.

3rd Semester		
Paper 1 **Major** **(Core)**	Material Studies 1 L-T-P-C : 4-0-0-4 Credits : 4 Scheme of Evaluation : Theory	Subject Code: PRD082M301 Level : 200

Course Objective: To develop understanding of materials, their properties and selection for product design applications.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand the classification and properties of materials.	BT1
CO 2	Apply material selection principles to design.	BT2
CO 3	Analyze materials for performance and sustainability.	BT3
CO 4	Select appropriate materials for products.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	**P**
I	**Introduction to Materials** Metals, polymers, ceramics, composites Material properties Structure-property relations	15	0
II	**Material Behaviour** Mechanical properties Thermal and electrical properties Degradation and durability	15	0
III	**Material Selection** Selection criteria	15	0

	Ashby charts Sustainability of materials		
IV	**Material Study Project** Material investigation Sample study Documentation	15	0
	TOTAL	**60**	**0**

Notional Credit Hours for the course: 30 x 4 = 120

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
4	60 hours	0 hours	8 hours
			Material Library, Testing, Sample Boards

Text Books:

Ashby, M. & Johnson, K. (2013); Materials and Design; Butterworth-Heinemann.

Reference Books:

Lefteri, C. (2014); Materials for Design; Laurence King.

3rd Semester		
Paper 2 Major (Core)	Human Factors & Ergonomics L-T-P-C : 1-0-6-4 Credits : 4 Scheme of Evaluation : Studio	Subject Code: PRD082M312 Level : 200

Course Objective: To introduce human factors and ergonomics and to enable learners to design products that fit human capabilities, needs and safety.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand principles of ergonomics and human factors.	BT1
CO 2	Apply anthropometric and ergonomic data to design.	BT2
CO 3	Analyze products for usability and safety.	BT3
CO 4	Create ergonomically informed product solutions.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	P
I	**Ergonomics Foundations** Human factors overview Anthropometry Physiology and biomechanics	4	23
II	**Cognitive Ergonomics** Perception and cognition Displays and controls Human error	4	23

III	**Applied Ergonomics** Workspace and product ergonomics Usability and safety Universal design	4	22
IV	**Ergonomics Project** User study Ergonomic evaluation Design proposal	3	22
	TOTAL	**15**	**90**

Notional Credit Hours for the course: 30 x 4 = 120

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
4	15 hours	90 hours	8 hours
			User Study, Measurement Labs, Project

Text Books:

Pheasant, S. & Haslegrave, C.M. (2006); Bodyspace; CRC Press.

Reference Books:

Tilley, A.R. (2002); The Measure of Man and Woman; Wiley.

3rd Semester		
Paper 3 **Minor**	Form Generation L-T-P-C : 1-0-6-4 Credits : 4 Scheme of Evaluation : Studio	Subject Code: PRD082N311 Level : 200

Course Objective: To develop the ability to generate, explore and refine three-dimensional product forms through systematic and intuitive methods.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand principles of form generation.	BT1
CO 2	Apply methods to generate and refine forms.	BT2
CO 3	Analyze forms for aesthetics and function.	BT3
CO 4	Create resolved product forms.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	P
I	**Form Fundamentals** Form vocabulary Proportion and balance Form and function	4	23
II	**Form Exploration** Transformation methods	4	23

	Form inspiration sources Rapid model making		
III	**Refinement** Surface and detail Semantics of form Form evaluation	4	22
IV	**Form Project** Form development Model making Presentation	3	22
	TOTAL	**15**	**90**

Notional Credit Hours for the course: 30 x 4 = 120

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
4	15 hours	90 hours	8 hours
			Model Making Studio, Reviews

Text Books:

Ashby, M. & Johnson, K. (2013); Materials and Design; Butterworth-Heinemann.

Reference Books:

Lidwell, W. & Manacsa, G. (2009); Deconstructing Product Design; Rockport.

3rd Semester		
Paper 4 **IDC**	North East Art and Craft L-T-P-C : 1-0-4-3 Credits : 3 Scheme of Evaluation : Studio	Subject Code: PRD082I311 Level : 200

Course Objective: To introduce learners to the rich and diverse art and craft traditions of North-East India, their materials, techniques and cultural context, and their potential in contemporary design.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand the art and craft traditions of North-East India.	BT1
CO 2	Apply knowledge of regional crafts to design.	BT2
CO 3	Analyze crafts in terms of material, technique and culture.	BT3
CO 4	Document and reinterpret North-East crafts for design.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	**P**
I	**Cultural Context** Peoples and cultures of North-East India Craft and community Symbolism and identity	4	15

II	**Textiles and Weaving** Handloom traditions Motifs and patterns Natural dyes	4	15
III	**Bamboo, Cane and Material Crafts** Bamboo and cane craft Pottery and wood craft Tools and techniques	4	15
IV	**Documentation and Design** Field documentation Craft mapping Contemporary application	3	15
	TOTAL	**15**	**60**

Notional Credit Hours for the course: 30 x 3 = 90

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
3	15 hours	60 hours	8 hours
			Craft Field Visits, Documentation, Presentations

Text Books:

Jaitly, J. (2012); Crafts Atlas of India; Niyogi Books.

Reference Books:

Ranjan, M.P. & Ranjan, A. (2007); Handmade in India; COHANDS.

3rd Semester		
Paper 5 **AEC**	Fundamentals of Business Communication L-T-P-C : 1-0-0-1 Credits : 1 Scheme of Evaluation : Theory	Subject Code: CEN982A301 Level : 200

Course Objective: To develop essential business communication skills enabling learners to communicate professionally in workplace and design-industry contexts.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand principles and channels of business communication.	BT1
CO 2	Apply formats of business correspondence and reports.	BT2
CO 3	Analyze professional communication situations.	BT3
CO 4	Demonstrate effective business writing and speaking.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	**P**
I	**Business Communication Basics** Importance and process	4	0

	Formal vs informal Communication networks		
II	**Business Correspondence** Business letters Emails and memos Notices and circulars	4	0
III	**Report Writing** Types of reports Structure Proposals	4	0
IV	**Professional Speaking** Meetings and minutes Interviews Presentations	3	0
	TOTAL	**15**	**0**

Notional Credit Hours for the course: 30 x 1 = 30

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
1	15 hours	0 hours	8 hours
			Mock Interviews, Business Writing Tasks, Presentations

Text Books:

Lesikar, R.V. & Flatley, M.E. (2009); Basic Business Communication; Tata McGraw Hill.

Reference Books:

Raman, M. & Singh, P. (2012); Business Communication; Oxford University Press.

3rd Semester		
Paper 6 **AEC**	Behvioural Sciences-III L-T-P-C : 1-0-0-1 Credits : 1 Scheme of Evaluation : Theory	Subject Code: BHS982A304 Level : 200

Course Objective: The objective of Behavioural Science-III is to develop the learner's self-awareness, interpersonal effectiveness and professional behaviour essential for a collaborative design practice.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand the key concepts of Behavioural Science-III.	BT1
CO 2	Apply behavioural skills in academic and studio settings.	BT2
CO 3	Analyze interpersonal and group situations effectively.	BT3
CO 4	Demonstrate professional and ethical conduct.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	**P**

I	**Leadership** Leadership styles Situational leadership Ethics in leadership	4	0
II	**Decision Making** Rational and intuitive models Problem solving Critical thinking	4	0
III	**Managing Change** Resistance to change Adaptability Resilience	4	0
IV	**Creativity and Innovation** Creative thinking tools Ideation Design mindset	3	0
	TOTAL	**15**	**0**

Notional Credit Hours for the course: 30 x 1 = 30

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
1	15 hours	0 hours	8 hours
			Role Plays, Group Activities, Self-assessment, Reflective Journaling

Text Books:

Robbins, S.P. & Judge, T.A. (2019); Organizational Behavior, 18th Ed.; Pearson.

Reference Books:

Goleman, D. (1995); Emotional Intelligence; Bantam Books.

3rd Semester		
Paper 7 **SEC**	Form Study L-T-P-C : 1-0-4-3 Credits : 3 Scheme of Evaluation : Studio	Subject Code: PRD082S311 Level : 200

Course Objective: To develop learners' understanding of three-dimensional form through hands-on study of proportion, surface and volume.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand the language of three-dimensional form.	BT1
CO 2	Apply form study methods through modeling.	BT2
CO 3	Analyze forms for proportion and surface quality.	BT3
CO 4	Create refined form studies.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	P
I	**Basics of Form** Primary solids Proportion and scale Positive and negative space	4	15
II	**Surface and Volume** Surface transitions Fillets and transitions Volume manipulation	4	15
III	**Form Language** Product semantics Style and character References	4	15
IV	**Form Study Project** Model series Refinement Presentation	3	15
	TOTAL	**15**	**60**

Notional Credit Hours for the course: 30 x 3 = 90

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
3	15 hours	60 hours	8 hours
			Model Making, Studio Reviews

Text Books:

Wong, W. (1993); Principles of Form and Design; Wiley.

Reference Books:

Lidwell, W. & Manacsa, G. (2009); Deconstructing Product Design; Rockport.

4th Semester		
Paper 1 Major (Core)	Advanced Product Design and CAD Application L-T-P-C : 1-0-6-4 Credits : 4 Scheme of Evaluation : Studio	Subject Code: PRD082M411 Level : 200

Course Objective: To advance product design capability through complex projects supported by computer-aided design (CAD) modeling and surfacing.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand advanced product design processes.	BT1
CO 2	Apply CAD modeling and surfacing to product design.	BT2
CO 3	Analyze design intent and manufacturability.	BT3
CO 4	Create CAD-developed product solutions.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	P
I	**Advanced Design Process** Complex brief framing Research and benchmarking Concept development	4	23
II	**CAD Modeling** Parametric solid modeling Assemblies Detailing and drawings	4	23
III	**Surfacing and Visualization** Surface modeling Rendering Design for manufacture basics	4	22
IV	**CAD Project** End-to-end product CAD documentation Presentation	3	22
	TOTAL	**15**	**90**

Notional Credit Hours for the course: 30 x 4 = 120

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
4	15 hours	90 hours	8 hours
			CAD Lab, Product Studio, Reviews

Text Books:

Ashby, M. & Johnson, K. (2013); Materials and Design; Butterworth-Heinemann.

Reference Books:

Shih, R.H. (2020); Parametric Modeling with SolidWorks; SDC Publications.

4th Semester		
Paper 2 Major (Core)	Packaging Design L-T-P-C : 1-0-6-4 Credits : 4 Scheme of Evaluation : Studio	Subject Code: PRD082M412 Level : 200

Course Objective: To enable learners to design packaging that protects, communicates and sells a product while addressing sustainability and manufacturability.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand the functions and constraints of packaging.	BT1
CO 2	Apply structural and graphic design to packaging.	BT2
CO 3	Analyze packaging for usability and sustainability.	BT3
CO 4	Create a complete packaging design solution.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	P
I	**Packaging Fundamentals** Roles of packaging Materials and manufacturing Regulatory and labelling requirements	4	23
II	**Structural Packaging** Dielines and nets Structural prototyping Ergonomics of packaging	4	23
III	**Graphics and Branding** Surface graphics Brand integration Shelf impact	4	22
IV	**Packaging Project** Concept to prototype Sustainability considerations Presentation	3	22
	TOTAL	**15**	**90**

Notional Credit Hours for the course: 30 x 4 = 120

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
4	15 hours	90 hours	8 hours
			Prototyping, Dieline Making, Studio Project

Text Books:

Roth, L. & Wybenga, G.L. (2005); The Packaging Designer's Book of Patterns; Wiley.

Reference Books:

DuPuis, S. & Silva, J. (2011); Package Design Workbook; Rockport.

4th Semester		
Paper 3 Major (Core)	Manufacturing Processes - 1 L-T-P-C : 4-0-0-4 Credits : 4 Scheme of Evaluation : Theory	Subject Code: PRD082M403 Level : 200

Course Objective: To provide foundational knowledge of manufacturing processes and their implications for product design.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand fundamental manufacturing processes.	BT1
CO 2	Apply process knowledge to design decisions.	BT2
CO 3	Analyze processes for cost, quality and suitability.	BT3

CO 4	Select appropriate processes for a product.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	P
I	**Introduction to Manufacturing** Role of manufacturing in design Classification of processes Materials and manufacturing	15	0
II	**Forming and Casting** Casting processes Forming and forging Sheet metal processes	15	0
III	**Machining** Turning, milling, drilling Tolerances and finish Machining considerations	15	0
IV	**Process Study** Process selection exercise Costing basics Documentation	15	0
	TOTAL	**60**	**0**

Notional Credit Hours for the course: 30 x 4 = 120

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
4	60 hours	0 hours	8 hours
			Workshop Visits, Process Study, Assignments

Text Books:

Kalpakjian, S. & Schmid, S. (2014); Manufacturing Engineering and Technology; Pearson.

Reference Books:

Groover, M.P. (2010); Fundamentals of Modern Manufacturing; Wiley.

4th Semester		
Paper 4 **Minor**	Product Analysis L-T-P-C : 1-0-4-3 Credits : 3 Scheme of Evaluation : Studio	Subject Code: PRD082N411 Level : 200

Course Objective: To develop critical skills in analyzing existing products in terms of form, function, usability, material and market.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand frameworks for product analysis.	BT1

CO 2	Apply analytical methods to existing products.	BT2
CO 3	Analyze products across multiple dimensions.	BT3
CO 4	Produce insightful product analysis reports.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	P
I	**Analysis Frameworks** Form and function analysis Usability analysis Material and construction analysis	4	15
II	**Teardown and Benchmarking** Product teardown Benchmarking Cost analysis basics	4	15
III	**User and Market** User research Market positioning Competitor analysis	4	15
IV	**Analysis Project** Comprehensive product analysis Insights and opportunities Presentation	3	15
	TOTAL	**15**	**60**

Notional Credit Hours for the course: 30 x 3 = 90

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
3	15 hours	60 hours	8 hours
			Product Teardown, Field Study, Report

Text Books:

Ulrich, K.T. & Eppinger, S.D. (2015); Product Design and Development; McGraw Hill.

Reference Books:

Cross, N. (2008); Engineering Design Methods; Wiley.

4th Semester		
Paper 5 **Minor**	Material Studies L-T-P-C : 3-0-0-3 Credits : 3 Scheme of Evaluation : Theory	Subject Code: PRD082N402 Level : 200

Course Objective: To develop understanding of materials, their properties and selection for product design applications.

Course Outcome:

After successful completion of the course, students will be able to		
CO	**Course Outcome**	**Blooms Taxonomy**

Level		Level
CO 1	Understand the classification and properties of materials.	BT1
CO 2	Apply material selection principles to design.	BT2
CO 3	Analyze materials for performance and sustainability.	BT3
CO 4	Select appropriate materials for products.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	P
I	**Introduction to Materials** Metals, polymers, ceramics, composites Material properties Structure-property relations	12	0
II	**Material Behaviour** Mechanical properties Thermal and electrical properties Degradation and durability	11	0
III	**Material Selection** Selection criteria Ashby charts Sustainability of materials	11	0
IV	**Material Study Project** Material investigation Sample study Documentation	11	0
	TOTAL	**45**	**0**

Notional Credit Hours for the course: 30 x 3 = 90

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
3	45 hours	0 hours	8 hours
			Material Library, Testing, Sample Boards

Text Books:

Ashby, M. & Johnson, K. (2013); Materials and Design; Butterworth-Heinemann.

Reference Books:

Lefteri, C. (2014); Materials for Design; Laurence King.

4th Semester		
Paper 6 AEC	Employability and Communication L-T-P-C : 2-0-0-1 Credits : 1 Scheme of Evaluation : Theory	Subject Code: CEN982A401 Level : 200

Course Objective: To enhance employability and maximize the learner's potential through advanced communication, workplace and career-readiness skills.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand the requirements of the professional workplace.	BT1
CO 2	Apply advanced written and oral communication for careers.	BT2
CO 3	Analyze and prepare effective resumes and portfolios.	BT3
CO 4	Demonstrate interview and group discussion competence.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	**P**
I	**Career Skills** Self-appraisal Goal setting Personal branding	8	0
II	**Resume and Cover Letters** Resume formats Cover letters Digital profiles	8	0
III	**Interview Skills** Types of interviews FAQs and answers Body language	7	0
IV	**Group Discussion and Presentation** GD techniques Presentation delivery Professional etiquette	7	0
	TOTAL	**30**	**0**

Notional Credit Hours for the course: 30 x 1 = 30

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
1	30 hours	0 hours	8 hours
			Mock Interviews, GD Rounds, Resume Building

Text Books:

Lesikar, R.V. & Flatley, M.E. (2009); Basic Business Communication; Tata McGraw Hill.

Reference Books:

Barun, K. Mitra (2011); Personality Development and Soft Skills; Oxford University Press.

4th Semester		
Paper 7 AEC	Behavioural Sciences-IV L-T-P-C : 1-0-0-1 Credits : 1 Scheme of Evaluation : Theory	Subject Code: BHS982A404 Level : 200

Course Objective: The objective of Behavioural Science-IV is to develop the learner's self-awareness, interpersonal effectiveness and professional behaviour essential for a collaborative design practice.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand the key concepts of Behavioural Science-IV.	BT1
CO 2	Apply behavioural skills in academic and studio settings.	BT2
CO 3	Analyze interpersonal and group situations effectively.	BT3
CO 4	Demonstrate professional and ethical conduct.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	**P**
I	**Professional Ethics** Values and integrity Workplace ethics Corporate social responsibility	4	0
II	**Stress and Well-being** Work-life balance Coping strategies Mindfulness	4	0
III	**Career Readiness** Professional conduct Networking Personal branding	4	0
IV	**Social Responsibility** Community engagement Sustainability mindset Civic values	3	0
	TOTAL	**15**	**0**

Notional Credit Hours for the course: 30 x 1 = 30

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
1	15 hours	0 hours	8 hours
			Role Plays, Group Activities, Self-assessment, Reflective Journaling

Text Books:

Robbins, S.P. & Judge, T.A. (2019); Organizational Behavior, 18th Ed.; Pearson.

Reference Books:

Goleman, D. (1995); Emotional Intelligence; Bantam Books.

5th Semester		
Paper 1 Major (Core)	User Interface Graphics L-T-P-C : 1-0-6-4 Credits : 4 Scheme of Evaluation : Studio	Subject Code: PRD082M511 Level : 300

Course Objective: To develop the skills required to design clear, consistent and aesthetically effective graphical user interfaces for digital products.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand principles of graphical user interface design.	BT1
CO 2	Apply visual design to interface elements.	BT2
CO 3	Analyze interfaces for consistency and clarity.	BT3
CO 4	Create UI screens and design systems.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	**P**
I	**UI Foundations** GUI principles Visual hierarchy Grids and layout for UI	4	23
II	**UI Components** Buttons, forms and navigation Iconography Colour and typography for UI	4	23
III	**Design Systems** Component libraries Style guides Responsive design	4	22
IV	**UI Project** Interface design brief High-fidelity screens Presentation	3	22
	TOTAL	**15**	**90**

Notional Credit Hours for the course: 30 x 4 = 120

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
4	15 hours	90 hours	8 hours
			Design Tools Lab, UI Studio, Reviews

Text Books:

Tidwell, J. (2010); Designing Interfaces; O'Reilly.

Reference Books:

Krug, S. (2014); Don't Make Me Think; New Riders.

5th Semester		
Paper 2 Major	User Experience L-T-P-C : 1-0-6-4 Credits : 4	Subject Code: PRD082M512

(Core)	Scheme of Evaluation : Studio	Level : 300

Course Objective: To introduce the theory and practice of user experience design, enabling learners to design usable, useful and desirable digital products.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand the principles and process of UX design.	BT1
CO 2	Apply UX research and design methods.	BT2
CO 3	Analyze user needs and usability.	BT3
CO 4	Create UX deliverables from research to prototype.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	**P**
I	**UX Foundations** What is UX UX process and disciplines Human-centred design	4	23
II	**UX Research** User research methods Personas and journeys Information architecture	4	23
III	**Interaction Design** Wireframing Prototyping Usability testing	4	22
IV	**UX Project** End-to-end case study Testing and iteration Presentation	3	22
	TOTAL	**15**	**90**

Notional Credit Hours for the course: 30 x 4 = 120

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
4	15 hours	90 hours	8 hours
			Research Fieldwork, Prototyping, Usability Testing

Text Books:

Garrett, J.J. (2010); The Elements of User Experience; New Riders.

Reference Books:

Cooper, A. et al. (2014); About Face; Wiley.

Reading Materials:

Norman, D. (2013); The Design of Everyday Things; Basic Books.

5th Semester		
Paper 3 Major (Core)	Material Studies - 2 L-T-P-C : 1-0-6-4 Credits : 4 Scheme of Evaluation : Studio	Subject Code: PRD082M503 Level : 300

Course Objective: To advance the study of materials with emphasis on emerging materials, finishes and their application in product design.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand advanced and emerging materials.	BT1
CO 2	Apply knowledge of finishes and surface treatments.	BT2
CO 3	Analyze materials for specific product applications.	BT3
CO 4	Specify materials and finishes for products.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	**Topics (if applicable) & Course Contents**	**Periods/ Hours**	
		L	**P**
I	**Advanced Materials** Smart materials Bio-based and sustainable materials Emerging composites	4	23
II	**Finishes and Treatments** Surface finishing Coatings and treatments Colour and texture	4	23
III	**Material Application** Material-product fit Case studies Specification	4	22
IV	**Material Project** Applied material study Prototype Documentation	3	22
	TOTAL	**15**	**90**

Notional Credit Hours for the course: 30 x 4 = 120

Total Credits in the Paper	**Lecture/ Tutorial**	**Studio/Practical**	**Experiential Learning**
4	15 hours	90 hours	8 hours
			Material Experiments, Sample Making, Reviews

Text Books:

Lefteri, C. (2014); Materials for Design; Laurence King.

Reference Books:

Ashby, M. & Johnson, K. (2013); Materials and Design; Butterworth-Heinemann.

5th Semester		
Paper 4 Project	Sustainable Product Design Project L-T-P-C : 1-0-4-4 Credits : 4 Scheme of Evaluation : Studio	Subject Code: PRD082M521 Level : 300

Course Objective: To enable learners to undertake a project that applies sustainability and eco-design principles across the life cycle of a product.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand sustainability principles in product design.	BT1
CO 2	Apply life-cycle and eco-design thinking to a project.	BT2
CO 3	Analyze products for environmental and social impact.	BT3
CO 4	Create a resolved sustainable product design.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	**Topics (if applicable) & Course Contents**	**Periods/ Hours**	
		L	**P**
I	**Sustainability Framing** Sustainability and product design Life cycle thinking Project brief and goals	4	15
II	**Research and Concept** Material and impact research Eco-design strategies Concept development	4	15
III	**Development** Prototyping Testing for sustainability Iteration	4	15
IV	**Project Outcome** Final sustainable product Documentation Presentation	3	15
	TOTAL	**15**	**60**

Notional Credit Hours for the course: 30 x 4 = 120

Total Credits in the Paper	**Lecture/ Tutorial**	**Studio/Practical**	**Experiential Learning**
4	15 hours	60 hours	8 hours
			LCA Exercise, Prototyping, Reviews

Text Books:

Ceschin, F. & Gaziulusoy, I. (2019); Design for Sustainability; Routledge.

Reference Books:

McDonough, W. & Braungart, M. (2002); Cradle to Cradle; North Point Press.

5th Semester		
Paper 5 Minor	Manufacturing Processes L-T-P-C : 1-0-6-4 Credits : 4 Scheme of Evaluation : Studio	Subject Code: PRD082N511 Level : 300

Course Objective: To provide foundational knowledge of manufacturing processes and their implications for product design.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand fundamental manufacturing processes.	BT1
CO 2	Apply process knowledge to design decisions.	BT2
CO 3	Analyze processes for cost, quality and suitability.	BT3
CO 4	Select appropriate processes for a product.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	**P**
I	**Introduction to Manufacturing** Role of manufacturing in design Classification of processes Materials and manufacturing	4	23
II	**Forming and Casting** Casting processes Forming and forging Sheet metal processes	4	23
III	**Machining** Turning, milling, drilling Tolerances and finish Machining considerations	4	22
IV	**Process Study** Process selection exercise Costing basics Documentation	3	22
	TOTAL	**15**	**90**

Notional Credit Hours for the course: 30 x 4 = 120

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
4	15 hours	90 hours	8 hours
			Workshop Visits, Process

			Study, Assignments

Text Books:

Kalpakjian, S. & Schmid, S. (2014); Manufacturing Engineering and Technology; Pearson.

Reference Books:

Groover, M.P. (2010); Fundamentals of Modern Manufacturing; Wiley.

6th Semester		
Paper 1 Major (Core)	Manufacturing Processes - 2 L-T-P-C : 4-0-0-4 Credits : 4 Scheme of Evaluation : Theory	Subject Code: PRD082M601 Level : 300

Course Objective: To extend knowledge of advanced manufacturing processes, joining and assembly relevant to complex product design.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand advanced and modern manufacturing processes.	BT1
CO 2	Apply joining, molding and assembly knowledge to design.	BT2
CO 3	Analyze processes for complex products.	BT3
CO 4	Integrate manufacturing knowledge into product development.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	**Topics (if applicable) & Course Contents**	**Periods/ Hours**	
		L	**P**
I	**Polymer and Composite Processing** Injection and blow molding Thermoforming Composites	15	0
II	**Joining and Assembly** Welding and fastening Adhesive bonding Design for assembly	15	0
III	**Advanced Manufacturing** Additive manufacturing CNC and automation Rapid prototyping	15	0
IV	**Applied Project** Process planning for a product Prototyping Documentation	15	0
	TOTAL	**60**	**0**

Notional Credit Hours for the course: 30 x 4 = 120

Total Credits in the Paper	**Lecture/ Tutorial**	**Studio/Practical**	**Experiential Learning**

4	60 hours	0 hours	8 hours
			Prototyping Lab, Workshop, Project

Text Books:

Kalpakjian, S. & Schmid, S. (2014); Manufacturing Engineering and Technology; Pearson.

Reference Books:

Boothroyd, G. et al. (2011); Product Design for Manufacture and Assembly; CRC Press.

6th Semester		
Paper 2 Major (Core)	Research Methodology L-T-P-C : 1-0-6-4 Credits : 4 Scheme of Evaluation : Studio	Subject Code: PRD082M612 Level : 300

Course Objective: To introduce learners to the field of design research and enable them to establish a strong research foundation grounded in both theory and practice, developing logical reasoning and the ability to conduct structured, in-depth research on a chosen design topic.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Define key research concepts, types and approaches, and distinguish between studio, dissertation and design research.	BT1
CO 2	Demonstrate knowledge of research proposals and appropriate tools for data collection and analysis.	BT2
CO 3	Organize literature review and case studies to identify research gaps.	BT3
CO 4	Analyze and apply data interpretation techniques and prepare structured design-research documents.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	**P**
I	**Introduction to Research** Meaning and types of research Qualitative, quantitative and mixed approaches Difference between studio, dissertation and design research Formulating a research proposal: aim, objectives, scope, limitations Research question versus hypothesis	4	23
II	**Tools and Methods** Selection of research topic and database search Literature review and case studies Design research methods and visual	4	23

	observation Primary and secondary data; sampling and questionnaires Data collection, survey techniques and analysis		
III	**Presentation Techniques** Formats for presenting data, case studies and analysis Media appropriate to design communication Two-dimensional drawings, models and digital presentation	4	22
IV	**Report and Paper Writing** Techniques of report and paper writing Synopsis and abstract formulation Reporting design development from concept to solution Citation, referencing and bibliography using online tools Introduction to plagiarism and academic integrity	3	22
	TOTAL	**15**	**90**

Notional Credit Hours for the course: 30 x 4 = 120

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
4	15 hours	90 hours	8 hours
			Literature Study, Field Research, Report Writing, Presentations

Text Books:

Booth, W.C., Colomb, G.G. & Williams, J.M. (2003); The Craft of Research; University of Chicago Press.

Milton, A. & Rodgers, P. (2013); Research Methods for Product Design; Laurence King.

Reference Books:

Creswell, J.W. (2003); Research Design: Qualitative, Quantitative and Mixed Method Approaches; Sage.

Gray, C. & Malins, J. (2004); Visualizing Research; Ashgate.

6th Semester		
Paper 3 Major (Core)	Technically Complex Product L-T-P-C : 1-0-6-4 Credits : 4 Scheme of Evaluation : Studio	Subject Code: PRD082M613 Level : 300

Course Objective: To enable learners to design technically complex products integrating mechanisms, electronics, materials and manufacturing.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**

CO 1	Understand the design of technically complex products.	BT1
CO 2	Apply integration of mechanics, electronics and materials.	BT2
CO 3	Analyze complex products for feasibility.	BT3
CO 4	Create a resolved complex product design.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	P
I	**Complex Product Foundations** Systems in products Mechanisms and moving parts Electronics integration basics	4	23
II	**Engineering Design** Design for function Tolerances and fits Reliability and safety	4	23
III	**Integration** Material and manufacturing integration Assembly Testing	4	22
IV	**Complex Product Project** Design development Prototype Presentation	3	22
	TOTAL	**15**	**90**

Notional Credit Hours for the course: 30 x 4 = 120

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
4	15 hours	90 hours	8 hours
			Prototyping, Engineering Studio, Reviews

Text Books:

Ulrich, K.T. & Eppinger, S.D. (2015); Product Design and Development; McGraw Hill.

Reference Books:

Pahl, G. & Beitz, W. (2007); Engineering Design; Springer.

6th Semester		
Paper 4 Major (Core)	Socio – Cultural Design L-T-P-C : 1-0-6-4 Credits : 4 Scheme of Evaluation : Studio	Subject Code: PRD082M614 Level : 300

Course Objective: To sensitize learners to the social and cultural dimensions of design and to enable design responses that are inclusive, responsible and context-sensitive.

Course Outcome:

After successful completion of the course, students will be able to

CO Level	Course Outcome	Blooms Taxonomy Level
CO 1	Understand the socio-cultural context of design.	BT1
CO 2	Apply participatory and inclusive design methods.	BT2
CO 3	Analyze designs for social and cultural impact.	BT3
CO 4	Create socially responsible design solutions.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	P
I	**Design and Society** Culture, society and design Design for diversity Ethics and responsibility	4	23
II	**Participatory Methods** Co-design and empathy Field research Stakeholder engagement	4	23
III	**Design for Impact** Design for development Accessibility and inclusion Social innovation	4	22
IV	**Socio-Cultural Project** Community brief Design intervention Presentation	3	22
	TOTAL	**15**	**90**

Notional Credit Hours for the course: 30 x 4 = 120

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
4	15 hours	90 hours	8 hours
			Field Study, Community Engagement, Presentations

Text Books:

Papanek, V. (1985); Design for the Real World; Thames & Hudson.

Reference Books:

Margolin, V. (2002); The Politics of the Artificial; University of Chicago Press.

6th Semester		
Paper 5 **Minor**	Packaging Design L-T-P-C : 1-0-6-4 Credits : 4 Scheme of Evaluation : Studio	Subject Code: PRD082N611 Level : 300

Course Objective: To enable learners to design packaging that protects, communicates and sells a product while addressing sustainability and manufacturability.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand the functions and constraints of packaging.	BT1
CO 2	Apply structural and graphic design to packaging.	BT2
CO 3	Analyze packaging for usability and sustainability.	BT3
CO 4	Create a complete packaging design solution.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	**P**
I	**Packaging Fundamentals** Roles of packaging Materials and manufacturing Regulatory and labelling requirements	4	23
II	**Structural Packaging** Dielines and nets Structural prototyping Ergonomics of packaging	4	23
III	**Graphics and Branding** Surface graphics Brand integration Shelf impact	4	22
IV	**Packaging Project** Concept to prototype Sustainability considerations Presentation	3	22
	TOTAL	**15**	**90**

Notional Credit Hours for the course: 30 x 4 = 120

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
4	15 hours	90 hours	8 hours
			Prototyping, Dieline Making, Studio Project

Text Books:

Roth, L. & Wybenga, G.L. (2005); The Packaging Designer's Book of Patterns; Wiley.

Reference Books:

DuPuis, S. & Silva, J. (2011); Package Design Workbook; Rockport.

7th Semester		
Paper 1 Major (Core)	System Design L-T-P-C : 1-0-6-4 Credits : 4 Scheme of Evaluation : Studio	Subject Code: PRD082M711 Level : 400

Course Objective: To enable learners to approach complex design problems holistically using systems thinking and to design coherent product-service systems.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand systems thinking and its application in design.	BT1
CO 2	Apply systems methods to complex design problems.	BT2
CO 3	Analyze systems for relationships and leverage points.	BT3
CO 4	Create an integrated system design solution.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	**P**
I	**Systems Thinking** Systems and complexity Elements, interconnections, purpose Mapping systems	4	23
II	**Service and Systems Design** Product-service systems Service blueprints Touchpoints and journeys	4	23
III	**Design for Systems** Stakeholder ecosystems Feedback loops Scenarios	4	22
IV	**System Design Project** Complex problem framing System proposal Presentation	3	22
	TOTAL	**15**	**90**

Notional Credit Hours for the course: 30 x 4 = 120

Total Credits in the Paper	Lecture/ Tutorial	Studio/Practical	Experiential Learning
4	15 hours	90 hours	8 hours
			Systems Mapping, Studio Project, Reviews

Text Books:

Meadows, D. (2008); Thinking in Systems; Chelsea Green.

Reference Books:

Stickdorn, M. & Schneider, J. (2011); This is Service Design Thinking; Wiley.

7th Semester		
Paper 2 Major (Core)	Product Detailing L-T-P-C : 1-0-6-4 Credits : 4 Scheme of Evaluation : Studio	Subject Code: PRD082M712 Level : 400

Course Objective: To develop skills in detailing products for production, including parts, joints, tolerances and documentation.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand product detailing for manufacture.	BT1
CO 2	Apply detailing, tolerancing and documentation.	BT2
CO 3	Analyze parts and assemblies for production.	BT3
CO 4	Produce complete product detailing documentation.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	**Topics (if applicable) & Course Contents**	**Periods/ Hours**	
		L	**P**
I	**Detailing Fundamentals** Part detailing Joints and fasteners Tolerances and fits	4	23
II	**Assembly Detailing** Assembly design Design for assembly BOM and part lists	4	23
III	**Documentation** Production drawings Specifications Standards	4	22
IV	**Detailing Project** Detailed product package Prototype verification Presentation	3	22
	TOTAL	**15**	**90**

Notional Credit Hours for the course: 30 x 4 = 120

Total Credits in the Paper	**Lecture/ Tutorial**	**Studio/Practical**	**Experiential Learning**
4	15 hours	90 hours	8 hours
			CAD Lab, Detailing Studio, Reviews

Text Books:

Boothroyd, G. et al. (2011); Product Design for Manufacture and Assembly; CRC Press.

Reference Books:

Bhatt, N.D. (2018); Machine Drawing; Charotar Publishing.

7th Semester		
Paper 3 Project	Research Project L-T-P-C : 1-0-4-4 Credits : 4	Subject Code: PRD082M721

	Scheme of Evaluation : Studio	Level : 400

Course Objective: To enable learners to undertake an independent, structured research project on a chosen design topic, applying rigorous methodology.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand the process of design research.	BT1
CO 2	Apply research methods to a chosen topic.	BT2
CO 3	Analyze and interpret research data.	BT3
CO 4	Produce a structured research report.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	**Topics (if applicable) & Course Contents**	**Periods/ Hours**	
		L	**P**
I	**Research Framing** Topic selection Aims, objectives and questions Scope and limitations	4	15
II	**Literature and Methods** Literature review Research methodology Data collection tools	4	15
III	**Analysis** Data analysis Interpretation Findings	4	15
IV	**Reporting** Report structure Citation and referencing Presentation	3	15
	TOTAL	**15**	**60**

Notional Credit Hours for the course: 30 x 4 = 120

Total Credits in the Paper	**Lecture/ Tutorial**	**Studio/Practical**	**Experiential Learning**
4	15 hours	60 hours	8 hours
			Independent Research, Fieldwork, Report Writing

Text Books:

Booth, W.C., Colomb, G.G. & Williams, J.M. (2003); The Craft of Research; University of Chicago Press.

Reference Books:

Creswell, J.W. (2003); Research Design; Sage Publications.

7th Semester		
Paper 4 **Minor**	Design for Sustainability L-T-P-C : 1-0-6-4 Credits : 4 Scheme of Evaluation : Studio	Subject Code: PRD082N711 Level : 400

Course Objective: To enable learners to integrate sustainability across the product life cycle through eco-design principles and strategies.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand principles of sustainable and eco-design.	BT1
CO 2	Apply life-cycle thinking to product design.	BT2
CO 3	Analyze products for environmental impact.	BT3
CO 4	Create sustainable product design solutions.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	**Topics (if applicable) & Course Contents**	**Periods/ Hours**	
		L	**P**
I	**Sustainability Foundations** Sustainability and design Life cycle assessment Circular economy	4	23
II	**Eco-Design Strategies** Design for disassembly Material and energy efficiency Design for longevity	4	23
III	**Systems and Services** Product-service systems Sharing and reuse models Social sustainability	4	22
IV	**Sustainability Project** Eco-design brief Sustainable solution Presentation	3	22
	TOTAL	**15**	**90**

Notional Credit Hours for the course: 30 x 4 = 120

Total Credits in the Paper	**Lecture/ Tutorial**	**Studio/Practical**	**Experiential Learning**
4	15 hours	90 hours	8 hours
			LCA Exercise, Eco-Design Project, Reviews

Text Books:

McDonough, W. & Braungart, M. (2002); Cradle to Cradle; North Point Press.

Reference Books:

Ceschin, F. & Gaziulusoy, I. (2019); Design for Sustainability; Routledge.

8th Semester		
Paper 1 Major (Core)	Design for Environmental Awareness L-T-P-C : 1-0-6-4 Credits : 4 Scheme of Evaluation : Studio	Subject Code: PRD082M811 Level : 400

Course Objective: To build awareness of environmental issues and to enable learners to create design communication and solutions that promote sustainability.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand environmental issues relevant to design.	BT1
CO 2	Apply sustainable design principles.	BT2
CO 3	Analyze the environmental impact of design decisions.	BT3
CO 4	Create design responses promoting environmental awareness.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	**Topics (if applicable) & Course Contents**	**Periods/ Hours**	
		L	**P**
I	**Environment and Design** Environmental challenges Sustainability and design Life cycle thinking	4	23
II	**Sustainable Strategies** Materials and energy Waste and circularity Eco-communication	4	23
III	**Design for Change** Behaviour change design Campaigns for awareness Green branding	4	22
IV	**Environmental Project** Sustainability brief Design solution Presentation	3	22
	TOTAL	**15**	**90**

Notional Credit Hours for the course: 30 x 4 = 120

Total Credits in the Paper	**Lecture/ Tutorial**	**Studio/Practical**	**Experiential Learning**
4	15 hours	90 hours	8 hours
			Field Study, Awareness Campaign, Presentations

Text Books:

McDonough, W. & Braungart, M. (2002); Cradle to Cradle; North Point Press.

Reference Books:

Thackara, J. (2005); In the Bubble: Designing in a Complex World; MIT Press.

8th Semester		
Paper 2 Major (Core)	Portfolio Design L-T-P-C : 1-0-6-4 Credits : 4 Scheme of Evaluation : Studio	Subject Code: PRD082M812 Level : 400

Course Objective: To enable learners to curate, design and present a professional portfolio that effectively communicates their skills and design identity.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand the purpose and formats of a design portfolio.	BT1
CO 2	Apply curation and layout to portfolio design.	BT2
CO 3	Analyze portfolios for narrative and impact.	BT3
CO 4	Create a professional print and digital portfolio.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	**Topics (if applicable) & Course Contents**	**Periods/ Hours**	
		L	**P**
I	**Portfolio Foundations** Purpose and audience Selecting and curating work Personal brand	4	23
II	**Portfolio Design** Layout and sequencing Case study storytelling Print and digital formats	4	23
III	**Self-Presentation** Personal identity system Online presence Interview presentation	4	22
IV	**Portfolio Project** Portfolio build Peer review Final presentation	3	22
	TOTAL	**15**	**90**

Notional Credit Hours for the course: 30 x 4 = 120

Total Credits in the Paper	**Lecture/ Tutorial**	**Studio/Practical**	**Experiential Learning**
4	15 hours	90 hours	8 hours
			Portfolio Studio, Mock Interviews, Reviews

Text Books:

Taylor, F. (2013); How to Create a Portfolio and Get Hired; Laurence King.

Reference Books:

Eisenman, S. (2008); Building Design Portfolios; Rockport.

8th Semester		
Paper 3 Dissertation	Dissertation/Research Project L-T-P-C : 1-0-4-8 Credits : 8 Scheme of Evaluation : Studio	Subject Code: PRD082M821 Level : 400

Course Objective: To enable learners to synthesize their learning into a substantial independent dissertation or major design project demonstrating research and design competence.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand the requirements of a design dissertation.	BT1
CO 2	Apply research and design methods independently.	BT2
CO 3	Analyze and develop a sustained body of work.	BT3
CO 4	Produce and defend a dissertation/major project.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	**Topics (if applicable) & Course Contents**	**Periods/ Hours**	
		L	**P**
I	**Proposal and Planning** Topic and rationale Research proposal Work plan	4	15
II	**Research and Development** Literature and case studies Concept development Iteration	4	15
III	**Resolution** Design resolution Testing and refinement Documentation	4	15
IV	**Defence** Dissertation report Final presentation Viva voce	3	15
	TOTAL	**15**	**60**

Notional Credit Hours for the course: 30 x 8 = 240

Total Credits in the Paper	**Lecture/ Tutorial**	**Studio/Practical**	**Experiential Learning**
8	15 hours	60 hours	8 hours
			Independent Study, Design

			Development, Viva

Text Books:

Booth, W.C., Colomb, G.G. & Williams, J.M. (2003); The Craft of Research; University of Chicago Press.

Reference Books:

Gray, C. & Malins, J. (2004); Visualizing Research; Ashgate.

8th Semester		
Paper 4 Minor	Minor Project L-T-P-C : 1-0-6-4 Credits : 4 Scheme of Evaluation : Studio	Subject Code: PRD082N812 Level : 400

Course Objective: To enable learners to undertake an independent minor-track project that applies and consolidates the knowledge and skills gained in their chosen minor specialization.

Course Outcome:

After successful completion of the course, students will be able to		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand the scope of an independent minor project.	BT1
CO 2	Apply minor-specialization skills to a defined problem.	BT2
CO 3	Analyze and develop the project through iteration.	BT3
CO 4	Produce and present a resolved minor project.	BT4

Prerequisite:

No Prerequisites

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	**P**
I	**Project Framing** Identifying the project brief Objectives and scope Work plan	4	23
II	**Research and Concept** Background research Concept development Feasibility	4	23
III	**Development** Iterative development Refinement Documentation	4	22
IV	**Outcome** Final deliverable Report Presentation	3	22
	TOTAL	**15**	**90**

Notional Credit Hours for the course: 30 x 4 = 120

Total Credits in	Lecture/ Tutorial	Studio/Practical	Experiential Learning

the Paper			
4	15 hours	90 hours	8 hours
			Independent Project Work, Reviews, Presentation

Text Books:

Ambrose, G. & Harris, P. (2010); Design Thinking; AVA Publishing.

Reference Books:

Milton, A. & Rodgers, P. (2013); Research Methods for Product Design; Laurence King.